

James Qin

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EDUCATION

Northeastern University, Boston, MA

Expected May 2029

Candidate for Bachelor of Science in Mechanical Engineering and Computer Science | GPA: 3.607

Relevant Coursework: Robot Dynamics & Control, Algorithms & Data, Statics, Materials Science, Calculus 3, Discrete Structures, Thermodynamics, Differential Equations, Physics 1, Program and Design Fundamentals 1

Activities: Northeastern Mars Rover Team

SKILLS & INTERESTS

Programming: Python, C++, Micro Python

Software: CAD (SolidWorks, Onshape), Git, VS Code, AutoCAD, MATLAB, Arduino IDE

Prototyping & Hardware: 3D Printing, Soldering, ESP32, Raspberry Pi Pico, Arduino

Interests: Art (Drawing/Sketching), Rock Climbing

EXPERIENCE

Northeastern Mars Rover Team | *Software & Mechanical Team Member*

September 2025 – Present

- Developed UI robotic-arm macro pad functionality in TypeScript and React, processing button and slider inputs to select operating modes and communicate mode commands to a corresponding ROS node.
- Designed and fabricated custom mounts and retention components for rover electronics, including USB hubs, a PCB, and Ethernet cabling

EDR | *STEM Instructor, Brockton, MA*

June 2026

- Taught hands-on STEM courses in Arduino programming, 3D printing, mathematics, and science to 60 students in grades 6–8.
- Guided students through designing and troubleshooting 20 engineering projects while introducing core computer science and design concepts.
- Developed and adapted technical lessons to make engineering, math, and programming concepts accessible to students at varying experience levels.

PROJECTS

5-DOF Robotic Arm | *SolidWorks, 3D Printing, MATLAB, ESP32*

August 2026 – Present

- Designing and fabricating a 5-DOF robotic arm with custom geared joints, carbon-fiber linkages, and servo/DC motor actuation.
- Redesigning a previous iteration to now have lighter carbon-fiber linkages and repositioned motors closer to the base to reduce distal mass; added a counterweight to offset gravitational torque at the elbow joint.
- Incorporated a planetary gearbox for elbow roll, added wrist-pitch motion, and revised joint assemblies with ball bearings and shaft collars for shaft support and retention.
- Developing MATLAB-based joint-control and kinematics software for arm simulation and Arduino hardware communication.

Carnival Crane Machine | *Arduino, SolidWorks, 3D Printing*

Spring 2025

- Designed and fabricated an electromechanical crane game with motorized 2D gantry motion and joystick control for an expo at the Boston Children's Museum.
- Developed an object-oriented control architecture in Micro Python using a state machine to manage game logic and system behavior on a Raspberry-Pi-Pico.
- Reconfigured the system into an Arduino-controlled ball-transfer game following an electronics failure on demo day, delivering a functional system for the final demonstration.